Retail Customer Segmentation & Product Recommendations

Project Report

Author: Touseeq Danish

RFM analytics, K-Means customer clustering, and interpretable cluster-based product recommendations.

Abstract

This project studies customer behavior in retail transaction data using exploratory data analysis, Recency-Frequency-Monetary (RFM) metrics, K-Means clustering, and a cluster-based product recommendation baseline. The repository includes a reusable pipeline, notebook, dashboard, and tests.

1. Problem statement

Retailers can use transaction histories to understand customer engagement and purchasing value. The project groups customers with similar transaction patterns and surfaces products that are popular among customers in the same group.

2. Dataset

The workflow expects the UCI Online Retail dataset with invoice identifiers, product descriptions, quantities, invoice dates, unit prices, customer identifiers, and optional country/product codes. The dataset is not bundled with the repository, so no row counts or computed statistics are claimed in this report.

3. Data preparation

The pipeline standardizes column names and parses dates and numeric fields. Rows without customer IDs, descriptions, invoice identifiers, valid dates, positive quantities, or positive unit prices are excluded from the positive-sales analysis. Customer IDs are not imputed. Returns and cancellations should be examined separately if relevant.

4. RFM features

Recency is measured in days from a customer's latest purchase to one day after the latest transaction date in the dataset. Frequency counts distinct invoices. Monetary value sums quantity multiplied by unit price. The dataset-derived reference date supports reproducibility on a fixed dataset.

5. Customer segmentation

RFM metrics are transformed using log1p and standardized before K-Means. Five clusters is the default configurable starting point, not a proven optimum. Cluster labels are arbitrary identifiers and must be interpreted through customer counts and RFM profiles.

6. Product recommendation approach

The recommender ranks products within each cluster by the number of distinct customers who purchased each product, using total units sold as a tie-breaker. It can filter out products already purchased by a selected customer. This is a transparent baseline, not a transformer-based recommender or a validated predictive model.

7. Generated outputs

Output file	Contents
cleaned_transactions.csv	Valid positive-sales transactions
customer_segments.csv	RFM features and cluster assignment per customer
cluster_product_recommendations.csv	Ranked product candidates per cluster

8. Testing

Unit tests use a synthetic dataset to check cleaning, RFM calculations, clustering, recommendation output structure, and invalid-input handling. Run the tests with pytest -q.

9. Limitations and future work

Compare cluster counts using silhouette scores and business interpretability; evaluate recommendations on a time-based holdout using precision@k, recall@k, or MAP@k; analyze returns separately; compare the baseline with association rules or item-item collaborative filtering; and apply privacy controls before processing customer data in production.

10. Conclusion

This project provides an end-to-end foundation for retail customer segmentation and interpretable product recommendations. Final business conclusions should be written only after running the workflow on the selected dataset and reviewing the measured results.

References

UCI Machine Learning Repository, Online Retail dataset: https://archive.ics.uci.edu/dataset/352/online+retail

Scikit-learn documentation, K-Means clustering: https://scikit-learn.org/stable/modules/clustering.html#k-means